ONC chain: conflict with an oncoming vehicle

Derivation runs downward, from the collision (lagging SPI) back to the earliest observable signals

LAGGING SPI: ONC collision

{ No. of times an ONC collision occurs per 100k manoeuvres } $< T_{ONC}$

Metric: contact between the ego vehicle and an oncoming vehicle.

why

PROXIMATE CAUSE

The ego vehicle occupied the conflict area while an oncoming vehicle was closing on it, and no avoidance action succeeded in time.

reaction tier (last resort, nearest the loss)

INTERMEDIATE LEADING SPI: AEB_ONC

{ No. of times AEB_ONC is triggered per 100k manoeuvres } $< T_{AEB_ONC}$

Metric: hard emergency brake while a gated oncoming vehicle was within the collision horizon. Mitigation attempted, collision not yet realised.

decompose the proximate cause (kinematic precursor tier)

OR gate: either cause alone is sufficient

CAUSE A

The accepted gap was already insufficient when the manoeuvre began.
Planning subsystem.

CAUSE B

Closing speed exceeded what the turn arc allowed because of increased oncoming velocity, so the deceleration needed rose beyond normal braking.
Prediction subsystem.

SIGNAL A

TTC to the nearest closing oncoming vehicle, sampled each frame.

SIGNAL B

Deceleration required to stop short of the conflict area at the current speed, sampled each frame.

observability and independence check

LEADING SPI: GAP_TTC

{ No. of times GAP_TTC $< \theta_{GAP_TTC}$ per 100k manoeuvres } $< T_{GAP_TTC}$

Metric: minimum gap TTC over the manoeuvre.

LEADING SPI: REQ_DECEL

{ No. of times REQ_DECEL $> \theta_{REQ_DECEL}$ per 100k manoeuvres } $< T_{REQ_DECEL}$

Metric: maximum required deceleration over the manoeuvre.

ATTACH TO THE SAFETY CASE

GAP_TTC on G2.1 and REQ_DECEL on G2.2 as leading evidence; AEB_ONC as intermediate leading and the ONC collision as lagging evidence on G2, the oncoming-conflict goal.

VALIDATE

Temporal precedence is checked against the ONC collision only. Incidents are attributed by actor identity, so a pedestrian or a rear impact is never counted here. The metric is separated between uneventful and ONC-collision manoeuvres.

AEB_ONC attaches directly, independent of the kinematic tier